

ECMT676 Econometrics II
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Tentative Solutions to Problem 4

1. (12.3)

- (a) Notice the model can be rewritten as $\ln \hat{E}(y|z) = \hat{\theta}_1 + \hat{\theta}_2 \ln z_1 + \hat{\theta}_3 z_2$ and, therefore,

$$\frac{\partial \ln \hat{E}(y|z)}{\partial \ln z_1} = \hat{\theta}_2.$$

- (b) Since $\ln \hat{E}(y|z_1, z_2 + 1) - \ln \hat{E}(y|z_1, z_2) = \hat{\theta}_3$, we have

$$\hat{\theta}_3 = \ln \frac{\hat{E}(y|z_1, z_2 + 1)}{\hat{E}(y|z_1, z_2)} = \ln \left[\frac{\hat{E}(y|z_1, z_2 + 1) - \hat{E}(y|z_1, z_2)}{\hat{E}(y|z_1, z_2)} + 1 \right] \cong \frac{\hat{E}(y|z_1, z_2 + 1) - \hat{E}(y|z_1, z_2)}{\hat{E}(y|z_1, z_2)}.$$

Thus, the percentage change in $\hat{\theta}(y|z)$ is $100\hat{\theta}_3$.

- (c) We know that $\partial \hat{E}(y|z) / \partial z_2 = \hat{E}(y|z)(\hat{\theta}_3 + 2\hat{\theta}_4 z_2)$ and thus the partial effect of z_2 on $\hat{E}(y|z)$ becomes negative if and only if $z_2 \leq -\frac{\hat{\theta}_3}{2\hat{\theta}_4}$ which is a consistent estimator of the turning point.

2. (12.10) The conditional mean function is $m(x_i, n_i, \beta) = n_i p(x_i, \beta)$. So we would, as usual, minimize the sum of squared residuals $\sum (y_i - n_i p(x_i, \beta))^2$ with respect to β . This gives us the nonlinear least square estimator $\hat{\beta}$. Define the weights as $\hat{w}_i \equiv n_i p(x_i, \hat{\beta})[1 - p(x_i, \hat{\beta})]$. Then the weighted nonlinear least square estimator minimizes $\sum \hat{w}_i^{-1} [y_i - n_i p(x_i, \hat{\beta})]^2$.

3. (13.2)

- (a) First, we can write down the conditional log-likelihood function for observation i , $\ln f(y_i|x_i) = -2^{-1} \ln(2\pi) - 2^{-1} \ln \sigma^2 - 2^{-1} \{[y_i - m(x_i, \beta)]/\sigma\}^2$. Then

$$\begin{aligned} \hat{\beta} &= \arg \max_{\beta} \sum_i \ln f(\theta|y_i, x_i) = \arg \max_{\beta} \sum_i -\frac{1}{2} \ln(2\pi) - \frac{1}{2} \ln \sigma^2 - \frac{1}{2} \left[\frac{y_i - m(x_i, \beta)}{\sigma} \right]^2 \\ &= \arg \min_{\beta} [y_i - m(x_i, \beta)]^2. \end{aligned}$$

Thus, we can see that the CMLE of β_0 is the nonlinear least square estimator.

- (b)

$$\frac{\partial \ln f(\theta|y_i, x_i)}{\partial \beta} = \frac{y_i - m(x_i, \beta)}{\sigma^2} m_{\beta}(x_i, \beta) \quad (1)$$

$$\frac{\partial \ln f(\theta|y_i, x_i)}{\partial \sigma^2} = -\frac{1}{2\sigma^2} + \frac{1}{2\sigma^4} [y_i - m(x_i, \beta)]^2 \quad (2)$$

where $m_{\beta}(x_i, \beta) \equiv \frac{\partial m(x_i, \beta)}{\partial \beta}(x_i, \beta)$ is a $P \times 1$ vector. Thus,

$$E[s_i(\theta_0|x_i)] = E \left[\begin{array}{c} \frac{y_i - m(x_i, \beta_0)}{\sigma_0^2} m_{\beta_0}(x_i, \beta_0) \\ -\frac{1}{2\sigma_0^2} + \frac{1}{2\sigma_0^4} [y_i - m(x_i, \beta_0)]^2 \end{array} \middle| x_i \right] = \left[\begin{array}{c} \frac{m_{\beta}}{\sigma_0^2} E(y_i - m(x_i, \beta_0)|x_i) \\ -\frac{1}{2\sigma_0^2} + \frac{1}{2\sigma_0^4} E[y_i - m(x_i, \beta_0)|x_i]^2 \end{array} \right] = \mathbf{0}$$

Note that we only use the fact that $E(y_i|x_i) = m(x_i, \beta_0)$ and $\text{var}(y_i|x_i) = \sigma_0^2$ in showing this. In other words, only the first two conditional moments of y_i need to be correctly specified; nothing else about the normal distribution is used.

- (c) From equation (2), we have $\hat{\sigma}^2 = N^{-1} \sum_i [y_i - m(x_i, \hat{\beta})]^2$.
(d) The Hessian of the log-likelihood function with respect to θ is

$$H_i(\theta|x_i) = \begin{bmatrix} -\frac{1}{\sigma^2} m_{\beta\beta'}(x_i, \beta) + \frac{y_i - m(x_i, \beta)}{\sigma^2} m_{\beta}(x_i, \beta) m_{\beta}(x_i, \beta)' & -\frac{1}{2\sigma^3} [y_i - m(x_i, \beta) m_{\beta}(x_i, \beta)] \\ -\frac{1}{2\sigma^3} [y_i - m(x_i, \beta) m_{\beta}(x_i, \beta)]' & \frac{1}{2\sigma^4} - \frac{1}{\sigma^6} [y_i - m(x_i, \beta)]^2 \end{bmatrix}$$

where $m_{\beta\beta'}(x_i, \beta) \equiv \frac{\partial^2 m(x_i, \beta)}{\partial \beta \partial \beta'}(x_i, \beta)$ is a $P \times P$ matrix.

- (e) We first show that

$$-E(H_i(\theta_0|x_i)) = \begin{bmatrix} \frac{1}{\sigma_0^2} m_{\beta\beta'}(x_i, \beta_0) & \mathbf{0} \\ \mathbf{0}' & \frac{1}{2\sigma_0^2} \end{bmatrix}.$$

Next we can see that

$$E(s_i(\theta_0|x_i) s_i(\theta_0|x_i)') = E \begin{bmatrix} s_{i,1}(\theta_0|x_i) s_{i,1}(\theta_0|x_i)' & s_{i,1}(\theta_0|x_i) s_{i,2}(\theta_0|x_i)' \\ s_{i,2}(\theta_0|x_i) s_{i,1}(\theta_0|x_i)' & s_{i,2}(\theta_0|x_i)^2 \end{bmatrix} \bigg| x_i = \begin{bmatrix} \frac{1}{\sigma_0^2} m_{\beta\beta'}(x_i, \beta_0) & \mathbf{0} \\ \mathbf{0}' & \frac{1}{2\sigma_0^2} \end{bmatrix}.$$

where $s_{i,1}(\theta_0|x_i)$ is the equation (1) and $s_{i,2}(\theta_0|x_i)$ the equation (2).

- (f) Since $\sqrt{N}(\hat{\theta} - \theta) \xrightarrow{d} N(0, [E(s_i(\theta_0) s_i(\theta_0)')]^{-1})$, the asymptotic variance of $\sqrt{N}(\hat{\beta} - \beta)$ is $\sigma^2 E(m_{\beta} m_{\beta}')^{-1}$.
Then, we can propose an estimator for the asymptotic variance of $\hat{\beta}$ by

$$\hat{\Sigma} = \hat{\sigma}^2 \left(\sum \hat{m}_{\beta} \hat{m}_{\beta}' \right)^{-1}$$

and the asymptotic standard errors of $\hat{\beta}_k$ is $\hat{\Sigma}_{kk}$, where $\hat{\sigma}^2 = N^{-1} \sum [y_i - m(x_i, \hat{\beta})]^2$, $\hat{m}_{\beta} = \frac{\partial m(x_i, \beta)}{\partial \beta}(x_i, \hat{\beta})$ and $\hat{\Sigma}_{kk}$ is the k^{th} element on the diagonal of $\hat{\Sigma}$.